Gary (Yunchu) Feng

Machine Learning Engineer · Computer Vision, Self-Supervised Learning, Robotics

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EDUCATION

Georgia Institute of Technology May 2022

B.S. Computer Engineering, Atlanta, GA

EXPERIENCE

Machine Learning Engineer Sep 2024 – Present

Microsoft

- Shipped a cross-encoder reranking service delivering **+12% CTR** in A/B test, serving 500 RPS at 90 ms p50.
- Launched a news-intelligence agent that ingests daily sources, clusters and dedupes events, and generates ranked summaries.
- Built ScoreCard Agent, an automated A/B experiment health analyzer running 25+ statistical checks (SRM, guardrails, Simpson's paradox, novelty effects) that produces SHIP/NO-SHIP verdicts with full audit trails.
- Deployed a personality-based evaluation harness using Prometheus-2 (7B LLM) to score Copilot suggestion pills at scale, improving offline/online alignment by 8%.
- Developed retrieval-augmented generation systems integrating BM25 keyword search with vector search via LlamaIndex.
- Distilled large language models, optimizing Phi-2 into a BERT-based model for accelerated inference and domain specialization.
- Designed data pipelines in Azure ML for scalable, secure model training and deployment.

Software Engineer Jul 2022 – Sep 2024

Microsoft

- Designed and maintained a Kubernetes-based forecasting service (GA product) serving **15,000+ clients**, including Dynamics 365 CRM and Microsoft 365 Support Center.
- Built a multi-region recommendation platform on Azure and Docker handling **20,000+ daily active users**.
- Led a cross-functional team with data scientists and a partner dev team to refine model outputs using A/B insights.
- Provided on-call support, diagnosing and resolving production incidents to maintain uninterrupted service.

Software Engineer May 2020 – Aug 2020

Skeena Bioenergy Ltd., Vancouver, BC

- Developed a computer vision application for automated PDF-to-SQL data migration, streamlining internal data processing.
- Created interactive Power BI dashboards from operational production data and built Python automation scripts for the internal toolset.

PUBLICATIONS

Evolution of Humanoid Locomotion Control Science Robotics, 2026

Y. Gu, G. Shi, F. Shi, I-C. Chang, Y-J. Wang, Q. Cheng, Z. Olkin, I. Lopez-Sanchez, **Y. Feng**, J. Zhang, A. D. Ames, H. Su, K. Sreenath. *Science Robotics* 11(117). doi:10.1126/scirobotics.aed3973

RESEARCH PROJECTS

Anatomy-shaped Masking for OCT I-JEPA Mar 2026 – Present

- Self-supervised I-JEPA pretraining of ViT-B/16 on 600K retinal OCT slices for glaucoma classification on Harvard FairVision.
- Proposed anatomy-shaped connected mask targets reaching **0.8947 test AUC**, beating random-masking I-JEPA (0.8878) at every probe and training regime (frozen +0.011, $p<0.0005$).
- Isolated target shape from target area with a matched-budget random control and a MIRAGE-placed rectangle-envelope baseline; ran a 2x3 probe-architecture ablation with paired-bootstrap confidence intervals.

- Released pretrained encoders on Hugging Face; documented two metric traps (validation loss and representation diversity both rank masking strategies incorrectly).

CopilotWorldLab – Latent World Models for Manipulation Jul 2026 – Present

- Reproduced V-JEPA 2-AC as a frozen coarse manipulation controller planning to goal images via CEM model-predictive control in MuJoCo.
- Built a reproducible 500-rollout benchmark (5 tasks x 2 objects x 50 fixed scenarios) scored from hidden simulator state, after robosuite and ManiSkill both proved unrunnable on the target platform.
- Diagnosed a CUDA allocator cliff causing a 4.6x planning slowdown at high CEM sample counts; fixed by batch chunking with numerically identical output.
- Investigating the model's own predictive energy as a competence gate for handoff to a classical visual servo.

SLIViT for 3D OCT Glaucoma Classification Mar – Apr 2026

- Reproduced the SLIViT ConvNeXt + ViT architecture for binary glaucoma classification on 10K OCT volumes, reaching 0.869 test AUC.
- Trained with 4-GPU DDP and fp16; showed 32 slices per volume matches 64 at a fraction of the cost.

Object Permanence Detection Feb 2026

- Built a video pipeline detecting physically impossible events using SAM2 mask propagation, RT-DETR detection, and DINOv2 re-identification.
- Generated narrative event logs consumable by an LLM for violation detection across occlusion, containment, and disappearance.

MOT17 Multi-Object Tracking Jan 2026

- Trained YOLOv8 with ByteTrack on MOT17 pedestrian data, improving tracking accuracy 8% over base YOLOv8.

EARLIER ENGINEERING PROJECTS

- **iValet, Intelligent Parking Management** (Georgia Tech ECE 4872 senior design, 2022) – camera-based vacancy detection with image segmentation, PostgreSQL state tracking, and a React UI with path planning. Advised by Dr. Patricio Vela.
- **Gesture-Controlled Robot Car** (ECE 4180, 2021) – OpenCV and MediaPipe finger detection on Raspberry Pi 4, commands sent over WiFi to an Mbed LPC1768 driving DC motors through an H-bridge.
- **Self-Driving DC Motor Car** (2021) – OpenCV lane detection with Canny edges and Hough transform, PID steering control across a Raspberry Pi 4 and Arduino with a custom-soldered H-bridge.

SKILLS

Languages: Python, C/C++, C#, SQL

ML & CV: PyTorch, self-supervised learning (I-JEPA, V-JEPA), vision transformers, LLM distillation, RAG, LlamaIndex, A/B experimentation

Infrastructure: Azure, Azure ML, Kubernetes, Docker, distributed training (DDP), Unix

Robotics & Embedded: ROS 2, MuJoCo, Raspberry Pi, Arduino, Mbed, IoT